

Brain

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State

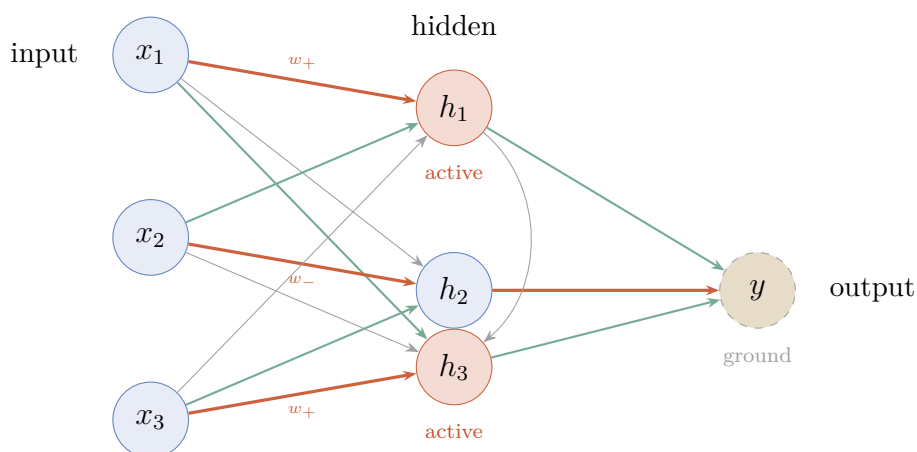
A brain is a weighted directed graph $G = (V, E, w)$ where each node $v \in V$ carries a state $x_v \in \mathbb{R}$ and each edge $(u, v) \in E$ carries a weight $w_{uv} \in \mathbb{R}$.

Update rule (discrete time):

$$x_v(t+1) = \sigma \left(\sum_{u \rightarrow v} w_{uv} x_u(t) + b_v \right)$$

where σ is a nonlinearity and b_v is a bias.

Ground state: $x^* = \sigma(Wx^* + b)$, a fixed point.



Ground

The network reaches ground when $\|x(t+1) - x(t)\| < \varepsilon$. Silence is the attractor. Activation is displacement from it.

Energy:

$$E(x) = -\frac{1}{2}x^\top Wx - b^\top x$$

Hopfield form. Decreases monotonically under asynchronous update when $W = W^\top$.

The Adaptive Connection

Each weight w_{uv} is adaptive:

$$w_{uv}(t+1) = w_{uv}(t) - \alpha \frac{\partial E}{\partial w_{uv}}$$

Large α : fast forgetting, near ground. Small α : memory persists. At $\alpha = 0$: frozen weights, fixed character.